

L3b: Lattice Models for Equity Pricing

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Objectives for Today

Today we build a binomial price model, estimate its parameters, and test it against the market patterns from L3a.

Today's concepts

- **Binomial lattice:** calculate the possible prices and their probabilities.
- **Model parameters:** estimate u , d , and p from data; calculate q for risk-neutral pricing.
- **Stylized facts:** explain why the fixed lattice matches weak autocorrelation but misses heavy tails and volatility clustering.

Lecture: [Lattice Models for Equity Pricing](#)

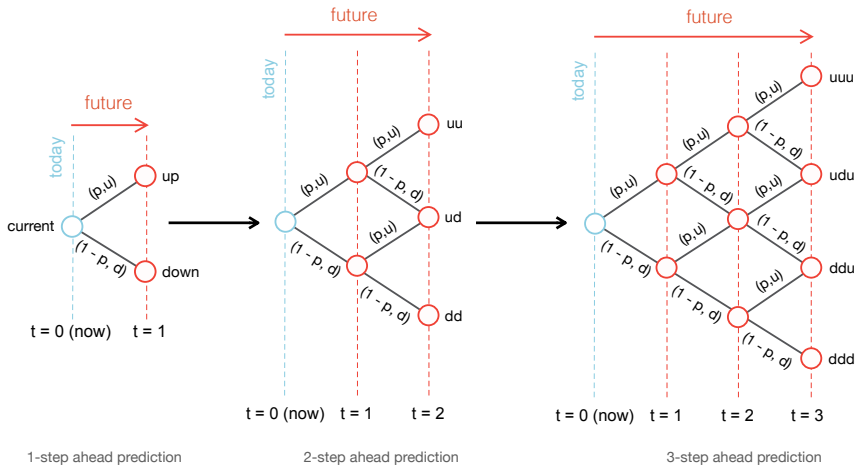
Lectures and Examples

Lecture: Lattice Models for Equity Pricing

Example: Binomial Lattice Simulation with Real-World Probability

- Check the implementation against a textbook example.
- Estimate the move factors and up probability from historical prices.
- Compare the model's predicted coverage with observed 2025 coverage.

A Recombining Price Lattice



Binomial Lattice Model

Let $S_0 > 0$ be today's share price and $\Delta t > 0$ the step length in years. The move factors satisfy $u > d > 0$, and the up probability is $0 < p < 1$. The share price after one step is given by:

$$S_1 = \begin{cases} S_0 u, & \text{with probability } p, \\ S_0 d, & \text{with probability } 1 - p. \end{cases}$$

The factors u , d , and probability p stay fixed. Successive moves are independent. After t steps, there are $t + 1$ possible prices.

Scope: we model share-price changes without dividend payments during the step.

Price Distribution

After t independent steps, let k be the number of up moves. There are $t - k$ down moves, so the price is given by:

$$\underbrace{S_{t,k} = S_0 u^k d^{t-k}}_{\text{price after } t \text{ steps}}, \quad k = 0, \dots, t.$$

There are $\binom{t}{k}$ ways to arrange those moves. Under the real-world probability rule $\mathbb{P}$, the node probability is given by:

$$\mathbb{P}(k \text{ up moves in } t \text{ steps}) = \underbrace{\binom{t}{k} p^k (1-p)^{t-k}}_{\text{binomial probability}}.$$

Different move orders can reach the same price.

Example: Recombination

The textbook check uses $S_0 = 20$, $u = 1.1$, $d = 0.9$, and $p = 0.6523$. An up move followed by a down move reaches the same price as the reverse.

This common price is given by:

$$\underbrace{20(1.1)(0.9)}_{\text{up then down}} = \underbrace{20(0.9)(1.1)}_{\text{down then up}} = 19.8.$$

Each path has probability $p(1 - p)$, so the shared node has probability $2p(1 - p)$.

This is **recombination**. It holds because $ud = du$. Here $ud = 0.99$, so recombination does not require $ud = 1$.

Optional extension: the recombination notebook compares path counts and node counts.

Estimate Parameters from Data

For forecasting, estimate (u, d, p) from growth rates $g_2, \dots, g_T$ observed at intervals of Δt years. Start two empty lists: up factors U and down factors D .

1. Append $e^{g_t \Delta t}$ to U if $g_t > 0$, or to D if $g_t < 0$. Skip zero-growth observations when building the lists.
2. Compute the average factors using:

$$u = \frac{1}{|U|} \sum_{v \in U} v, \quad d = \frac{1}{|D|} \sum_{v \in D} v.$$

3. Estimate the up probability as the fraction of observed steps that rise:

$$p = \frac{|U|}{T - 1}.$$

Counts: $|U|$ and $|D|$ are list lengths. Repeated factors count separately. Zero-growth days stay in the denominator of p ; both lists must be nonempty.

Check the Model Against New Data

Historical estimates give us a model for forecasting.

- The estimates depend on which observations we use.
- Two move factors summarize many different observed price changes.
- Test the fitted model on later data: compare its predicted price range with the prices that actually occur.

Example: the companion notebook compares model-predicted coverage with observed 2025 coverage.

Forecasting and Pricing

Forecasting asks which prices may occur. Pricing asks what a future payoff is worth today. These tasks use different probability rules:

Real-world $\mathbb{P}$

- Estimate the up probability p from data.
- Use it to forecast prices and measure risk.

Risk-neutral $\mathbb{Q}$

- Calculate q from the move factors and risk-free growth.
- Use it to value future payoffs.

The pricing weight q is not a forecast of the frequency of up moves.

No-Arbitrage Condition

An **arbitrage** costs nothing initially, cannot lose money, and makes money in at least one future state.

Let g_y be the risk-free growth rate in inverse years. Over Δt years, its accumulation factor is given by:

$$R_f = e^{g_y \Delta t}.$$

To avoid arbitrage, the factors must satisfy the condition:

$$\underbrace{d < R_f < u}_{\text{no-arbitrage condition}}.$$

- If $R_f \leq d$, borrow at the risk-free rate and buy the share.
- If $R_f \geq u$, sell a borrowed share and invest the proceeds.

Arbitrage: Two Examples

In each example, the share costs $S_0 = 100$ and the trade starts with no net cash investment.

Borrow and buy: $R_f = d = 1.05$, $u = 1.20$

- Borrow 100 and buy one share. The loan repayment is 105.
- Sell for 105 or 120. Profit after repayment: 0 or 15.

Sell short and invest: $R_f = u = 1.10$, $d = 0.90$

- Sell a borrowed share for 100 and invest it. The investment becomes 110.
- Buy back the share for 110 or 90. Profit: 0 or 20.

Model assumptions: borrowing, lending, and short selling are allowed; transaction costs are ignored.

Calculate the Risk-Neutral Probability

With arbitrage ruled out, choose the risk-neutral up probability q so the weighted average price factor equals R_f . With down probability $1 - q$, this condition is written as:

$$qu + (1 - q)d = R_f.$$

Solving this equation for q gives the pricing weight:

$$q = \underbrace{\frac{R_f - d}{u - d}}_{\text{risk-neutral up probability}}.$$

Since $d < R_f < u$, we have $0 < q < 1$.

Meaning: q is a pricing weight. The historical up probability remains p .

Value a Future Payoff

Suppose a claim pays H_u dollars after an up move and H_d dollars after a down move. Its no-arbitrage value today, V_0 , is given by:

$$V_0 = \underbrace{\frac{1}{R_f}}_{\text{discount factor}} \underbrace{[qH_u + (1 - q)H_d]}_{\text{weighted payoff}}.$$

1. Weight the two payoffs using the risk-neutral probability q .
2. Divide by R_f to discount back one step.

Later in the course: we derive this pricing rule when we study options.

Test the Lattice Against Market Data

We now have a pricing rule. But how well does the underlying model describe market data? Return to the three patterns from L3a:

- **Heavy tails:** large growth rates occur more often than a fitted Normal model predicts.
- **Weak linear autocorrelation:** growth rates on different days have little linear relationship beyond short lags.
- **Volatility clustering:** large movements tend to occur together, as do small movements.

Probability rule: use real-world $\mathbb{P}$ to compare the model with market data. See [Cont \(2001\)](#) for the empirical patterns.

Growth-Rate Distribution

To make that comparison, express the lattice moves as growth rates. At step j , the price factor is u or d . For a step of length Δt years, the growth rate is given by:

$$g_j = \frac{1}{\Delta t} \ln \left(\frac{S_j}{S_{j-1}} \right) = \begin{cases} \ln u / \Delta t, & \text{with probability } p, \\ \ln d / \Delta t, & \text{with probability } 1 - p. \end{cases}$$

The model produces only two possible growth rates, both in inverse years. Changing p changes how often each value occurs, not the values themselves.

One-Step Mean and Variance

Keep $u > d > 0$, $0 < p < 1$, and $\Delta t > 0$ years fixed.

The mean growth rate μ_g under $\mathbb{P}$ is given by:

$$\mu_g \equiv \mathbb{E}_{\mathbb{P}}[g_j] = \underbrace{\frac{p \ln u + (1 - p) \ln d}{\Delta t}}_{\text{mean growth rate}}.$$

The variance σ_g^2 is given by:

$$\sigma_g^2 \equiv \text{Var}_{\mathbb{P}}(g_j) = \underbrace{\frac{p(1 - p)}{(\Delta t)^2} \left[\ln\left(\frac{u}{d}\right) \right]^2}_{\text{growth-rate variance}}.$$

Both quantities stay fixed because u , d , and p stay fixed.

Heavy Tails

The two growth rates have a maximum magnitude. This bound, M , is given by:

$$M = \frac{\max\{|\ln d|, |\ln u|\}}{\Delta t}.$$

Since every one-step growth rate satisfies $|g_j| \leq M$, the probability of exceeding this bound is given by:

$$\underbrace{\mathbb{P}(|g_j| > x) = 0 \quad \text{for } x > M}_{\text{no observations beyond the bound}}.$$

The fixed lattice cannot produce heavy-tailed one-step growth rates.

Temporal Dependence

Each growth rate depends on its own up or down move. Since successive moves are independent, so are growth rates at different steps.

Let $\tau \geq 1$ be an integer lag. Their covariance is given by:

$$\text{Cov}_{\mathbb{P}}(g_j, g_{j+\tau}) = 0.$$

Since the growth-rate variance σ_g^2 is positive, the ACF is given by:

$$\underbrace{\rho_g(\tau) = \frac{\text{Cov}_{\mathbb{P}}(g_j, g_{j+\tau})}{\sigma_g^2}}_{\text{zero autocorrelation}} = 0.$$

Interpretation: the model matches weak growth-rate autocorrelation because we assumed independent moves.

Volatility Clustering

Independent moves also give independent growth-rate magnitudes. For any integer lag $\tau \geq 1$, their covariance is given by:

$$\text{Cov}_{\mathbb{P}}(|g_j|, |g_{j+\tau}|) = 0.$$

Let $\mathcal{F}_j$ represent the information in the first j moves. Given that history, the future growth-rate variance is still given by:

$$\underbrace{\text{Var}_{\mathbb{P}}(g_{j+\tau} \mid \mathcal{F}_j)}_{\text{constant conditional variance}} = \sigma_g^2 .$$

A large move today does not make a large move tomorrow more likely. The fixed lattice has no volatility clustering.

The Lattice Passes One of Three Tests

Compare the fixed lattice with the market patterns from L3a:

- **Heavy tails: fails.** Only two one-step growth rates are possible.
- **Weak growth-rate autocorrelation: passes.** Independent moves give zero autocorrelation.
- **Volatility clustering: fails.** Past movements do not change future variance.

The lattice is a useful baseline, but these assumptions limit its ability to describe market data.

Optional Advanced Material

These notebooks extend the main lesson:

- [Recombination and Computational Cost](#)
Compare path and node counts; distinguish repeated draws, recombination, and reciprocal factors.
- [Bootstrap Uncertainty in Lattice Calibration](#)
Measure how estimates of (u, d, p) change across resampled data.
- [Binomial Return Derivations](#)
Derive cumulative log returns, Normal limits, and correlations caused by overlapping windows.

Optional: none is required for L4a.

Summary

This lecture built a binomial lattice, estimated its parameters, and checked its limits against market data.

Key takeaways

- **Two moves build a price distribution.** The number of up moves determines the price at each step.
- **Forecast with p ; price with q .** Estimate (u, d, p) from data. Calculate q from u , d , and R_f .
- **The fixed lattice passes one of three tests.** It has zero growth-rate autocorrelation, but no heavy tails or volatility clustering.

Its fixed, independent moves make the lattice simple and explain its limits.

That's a wrap. Which lattice assumption would you change first, and which market pattern should the new model capture?